A Stacked Ensemble Machine Learning Technique for Robust Multi-Terrain Classification in Legged Robots

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ABSTRACT

The increasing adoption of legged robots in search and rescue, monitoring, and exploration demands accurate terrain classification for adaptive locomotion. However, existing methods struggle with generalizability and accuracy due to terrain variability and sensor noise. To address these challenges, this study proposes a multi-terrain classification framework leveraging ensemble machine-learning techniques. The framework combines Random Forest (RF), Support Vector Machine (SVM), and K-Nearest Neighbors (KNN) models, further integrated through a Stacking Ensemble (SE) learning to maximize classification performance. The system utilizes a force sensor and IMU data from a quadruped robot to extract meaningful features for robust classification. The Stacking Ensemble achieved the highest accuracy (0.9983), highlighting its superior classification performance despite increased training time compared to KNN (0.9844), SVM (0.9896), and RF (0.9757). This highlights the overall effectiveness of ensemble learning in handling the complexities of multi-terrain environments. Future work may explore real-time implementation and generalization to unseen terrains.

CCS CONCEPTS

• **Computing methodologies** → **Ensemble methods**; *Classification and regression trees*; • **Computer systems organization** → Robotic autonomy.

KEYWORDS

Terrain classification, legged robots, ensemble learning, sensor fusion, adaptive locomotion, machine learning

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1 INTRODUCTION

The advent of legged robots has revolutionized exploration and intervention in challenging environments where traditional wheeled or tracked vehicles struggle to operate effectively [8]. From search and rescue missions in disaster-stricken areas to planetary exploration on unstructured terrains, legged robots offer unprecedented mobility and adaptability. Unlike conventional robotic platforms, legged robots can navigate complex landscapes such as rubble, steep inclines, and irregular surfaces, making them critical for scenarios requiring precise locomotion and terrain negotiation. The biomimetic design of legged robots, inspired by biological locomotion systems, enables them to traverse diverse terrains with remarkable agility and resilience [12]. Moreover, these robots have significant potential in military surveillance, environmental monitoring, and humanitarian assistance, where terrain variability poses navigational challenges [15].

Machine learning techniques have emerged as powerful approaches for terrain classification in robotic systems, with various models demonstrating different capabilities and limitations [10]. Supervised learning algorithms, such as SVMs, RFs, and Neural Networks, have been extensively employed for terrain recognition [11]. Decision trees offer interpretable classifications, while ensemble methods like bagging and boosting can improve predictive accuracy by aggregating multiple weak learners [7]. Deep learning architectures, particularly Convolutional Neural Networks (CNNs), have shown promising results in extracting complex spatial features from sensor data [9]. Probabilistic models like Gaussian Processes provide uncertainty estimates alongside classifications, enhancing robotic decision-making [6]. Additionally, transfer learning techniques enable models to generalize across different robotic platforms and sensor configurations [3]. Hybrid approaches combining multiple classification techniques have demonstrated superior performance compared to single-model strategies [4]. These diverse classification models address the intricate challenge of accurately categorizing terrain types based on sensor measurements.

Existing research in terrain classification has primarily focused on leveraging various sensor modalities, including force sensors, Inertial Measurement Units (IMUs), visual sensors, and tactile feedback [11]. Early approaches relied on manual feature engineering and rule-based systems, which exhibited limited generalizability across different robotic platforms [17]. Some researchers have utilized force sensor data to distinguish terrain properties by analyzing ground reaction forces and pressure distributions [17]. IMU-based techniques exploit angular velocities and linear accelerations to infer terrain characteristics, demonstrating moderate success in controlled environments [11]. Computer vision methods employing texture analysis and machine learning have also been explored, though they suffer from computational complexity and sensitivity to lighting conditions [18]. Legacy classification models often struggle with several critical limitations: reduced accuracy in complex, unstructured environments, poor generalization across different robot morphologies, and high computational overhead [10]. The dynamic nature of terrain interactions necessitates adaptive classification techniques that can handle real-time sensor fusion and robust feature extraction [15]. The inherent variability in sensor measurements poses significant challenges for developing effective terrain classification frameworks. Furthermore, most existing approaches lack comprehensive validation across diverse terrain types, limiting their practical applicability in mission-critical scenarios [7].

To address the current limitations in robotic terrain recognition, this work introduces a multi-terrain classification framework for legged robots, designed to enhance adaptive locomotion by leveraging ensemble machine learning techniques. We combine RF, SVM, and KNN models, integrating them through a Stacking Ensemble learning approach to optimize classification performance. The study utilizes force sensors and IMU data from a quadruped robot to extract meaningful features for robust terrain classification. The results highlight the effectiveness of ensemble learning in handling the complexities of multi-terrain environments.

2 DATASET DETAILS

The QCAT legged robot dataset provides a comprehensive set of sensor measurements from the quadruped robot, DyRET, designed to analyze terrain interactions and locomotion efficiency across varied conditions [2][1]. A representative schematic is given in Fig. 1. The dataset incorporates two primary sensor streams: force sensors on the robot's four feet and an IMU (Inertial Measurement Unit) on its body. Force sensor data includes triaxial force readings (x, y, z components) for each leg, capturing the interaction dynamics between the robot and the ground. IMU data provides angular velocities, linear accelerations, and quaternion-based orientations to track the robot's body movement in 3D space. The dataset spans six distinct terrains (concrete, grass, gravel, mulch, dirt, and sand) and six speed settings, with ten trials per terrain-speed combination. Each trial consists of eight steps, yielding a total of 2,880 steps from 6 different terrains, six different speeds, ten trials, and eight steps. This extensive dataset, enabling diverse locomotion analysis, is instrumental for robotic perception, control, and terrain classification research. Each trial in the dataset contains approximately 5,300 samples with 8 steps, making it a valuable resource for machine learning applications. One step is represented

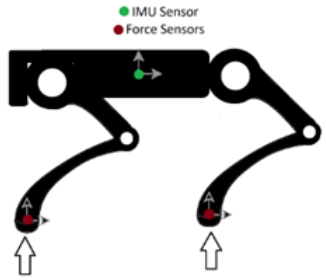

Figure 1: Quadruped robot with force and IMU sensor location (representative)

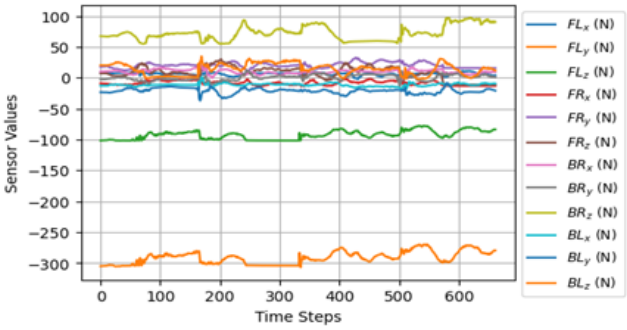

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Figure 2: Sample data visualization of the dataset of Force sensor readings for one step of the robot. The force sensors' values measure ground reaction forces (in N) on each leg (FL: Forward Left, FR: Forward Right, BR: Backward Right, BL: Backward Left) along x, y, z axes.

for visualization purposes in Fig. 2 for Force sensors and Fig. 3 for IMU sensors. Jupyter notebooks are used to preprocess data from CSV files for terrain classification or gait optimization tasks. The force sensor files record the dynamic forces experienced by each leg on different surfaces and speeds. Similarly, the IMU files offer insights into body dynamics under various locomotion conditions. Studies leveraging this dataset have enhanced understanding of quadruped locomotion, such as adaptive terrain navigation [17] and gait stabilization [12].

3 PROPOSED STACKED ENSEMBLE MODEL

The proposed methodology for multi-terrain classification is to design an ensemble learning model that involves a combination of machine learning models, each chosen for its specific strengths and ability to process complex sensor data. After conducting numerical experiments with different machine learning models, such as XGBoost, LightGBM, CatBoost, Logistic Regression, Ridge, GaussianNB, KNN, RF, SVM, and Decision Tree, the combination of RF, SVM, and KNN came out as the most suitable base learners as they offer the best accuracy with minimal computational time. To further enhance performance, these models are integrated into a SE learning (Fig. 4), which combines their predictions to achieve superior accuracy. Below, we define each model in detail and explain the mathematical framework underlying their operations.

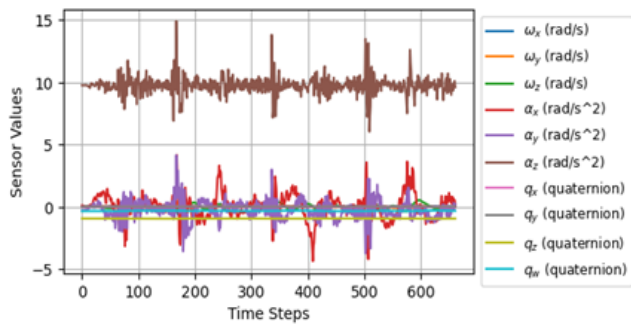

Figure 3: Sample data visualization of the dataset of IMU sensor readings for one step of the robot. The IMU sensors' values capture angular velocity, ω (in rad/s), linear acceleration, α (in m/s^2), and 3D orientation, q (in quaternion).

RF is an ensemble learning method that builds multiple decision trees during training and combines their outputs to make final predictions [5]. It achieves robustness and reduces the risk of overfitting by introducing randomness through the bagging technique, where each tree is trained on a random subset of the data. This approach has effectively analyzed terrain sensor datasets, as demonstrated in terrain classification studies [17]. Mathematically, a Random Forest is an ensemble of N decision trees, where the final prediction $f_1(x)$ is the majority vote (classification) or the mean (regression) of individual trees:

$$f_1(x) = \text{mode}\left(\{T_1(X), T_2(X), \ldots, T_N(X)\}\right) \text{ (classification)}$$
$$f_1(x) = \frac{1}{N}\sum_{i=1}^{N} T_i(X) \text{ (regression)} \tag{1}$$

where $T_i(X)$ is the prediction of the i-th decision tree.

SVM is a supervised learning algorithm designed to find the optimal hyperplane that separates data points of different classes in a feature space [16]. SVM ensures robust classification by maximizing the margin between the hyperplane and the nearest data points. It can also handle non-linearly separable data by mapping it to a higher-dimensional space using kernel functions, making it powerful in identifying complex patterns in datasets such as IMU and force sensor readings. Its application in robotic movement strategies highlights its adaptability to intricate sensor data [4]. Mathematically, the SVM objective is to maximize the margin between data points of different classes. The decision boundary is represented as:

$$f_2(x) = w^T x + b \tag{2}$$

where w is the weight vector, b is the bias term, x is the input vector. For non-linear kernels, the decision function becomes:

$$f_2(x) = \sum_{i=1}^{N} \alpha_i y_i K(x, x_i) + b \tag{3}$$

where $K(x, x_i)$ is the kernel function, α_i are Lagrange multipliers, y_i are the class labels.

KNN is a non-parametric, instance-based learning algorithm that classifies data based on the majority class of its K nearest neighbors [13]. It uses a distance metric, such as Euclidean or Manhattan distance, to determine similarity between data points. KNN requires no explicit training phase but relies on efficient search structures like KD-Trees or Ball Trees for scalability. Its ability to handle both numerical and categorical data makes it a versatile choice for terrain classification, particularly in scenarios where class boundaries are complex and non-linear. Mathematically, for classification, KNN assigns the most frequent label among neighbors:

$$f_3(x) = \arg\max_c \sum_{i \in N_K(x)} 1(y_i = c) \tag{4}$$

where $N_K(x)$ is the set of K nearest points, determined by a distance metric $d(x, x_i)$ such as Manhattan distance $(d(x, x_i))$.

$$d(x, x_i) = \sum_{j=1}^{n} \left|x_j - x_{i,j}\right| \tag{5}$$

Weighted KNN applies distance-based weighting:

$$w_i = \frac{1}{d(x, x_i) + \epsilon} \tag{6}$$

The SE learning combines the strengths of all these base models to improve classification performance further. By using the predictions of the RF, SVM, and KNN input features for a meta-learner, the ensemble captures the unique contributions of each model and leverages them to make more accurate final predictions. Logistic Regression is employed as the meta-learner in this project, providing a simple yet effective mechanism for combining the outputs of the base models. Mathematically, Stacking combines predictions from base models $\{f_1(x), f_2(x), \ldots, f_K(x)\}$ using a meta-model g :

$$\hat{y} = g\left(f_1(x), f_2(x), f_3(x)\right) \tag{7}$$

For classification, the meta-model g is often Logistic Regression:

$$\hat{y} = \text{sigmoid}\left(w^T \hat{F} + b\right) \tag{8}$$

where $\hat{F}$ is the concatenated output of base model predictions.

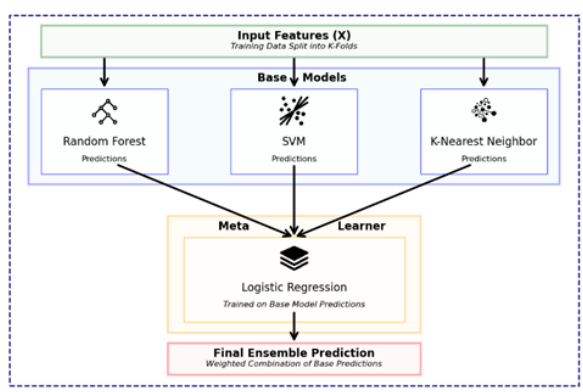

Figure 4: Proposed Stacked Ensemble architecture

4 PERFORMANCE METRICS

Performance metrics are essential to assess the machine learning models in terrain classification. We evaluate using accuracy, precision, recall, F1-score, and the confusion matrix to capture different aspects of performance. While accuracy reflects the overall correctness, it can be misleading with imbalanced data. Hence, additional metrics ensure the classifier performs reliably across all terrain types. Precision shows how well the model avoids false positives, recall reflects its ability to minimize false negatives, and the F1-score balances both. These metrics are crucial for distinguishing terrains with overlapping features like gravel and sand.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{9}$$

$$\text{Precision} = \frac{TP}{TP + FP} \tag{10}$$

$$\text{Recall} = \frac{TP}{TP + FN} \tag{11}$$

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{12}$$

In the above expressions, TP, TN, FP, and FN represent True Positives, True Negatives, False Positives, and False Negatives, respectively. The confusion matrix provides a detailed breakdown of the model's predictions for each class. It highlights the TP, TN, FN, and FP for each terrain type, offering a visual and quantitative understanding of classification errors. By analyzing the confusion matrix, we can identify specific terrains where the model may struggle, such as misclassifying grass as dirt due to similar sensor patterns.

5 RESULTS AND DISCUSSIONS

The contrast models (RF, SVM, KNN) were trained and evaluated based on accuracy, precision, recall, F1-score, and training time. Finally, the proposed SE learning model was also implemented and all performance metrics, along with confusion matrices, are presented below.

5.1 Performance Metrics

The Random Forest model alone achieved an accuracy of 0.9757 with a training time of 24.7444 seconds. The model exhibited strong classification performance across all classes, with precision ranging from 0.9159 to 1.0000. While the recall for most classes was close to 1.0000, class 3 showed a slightly lower recall of 0.9213, indicating some misclassification in this category. Overall, the model demonstrated high generalization capabilities with balanced precision and recall, which can be seen in the results given below in Table 1 and confusion matrix given below in Fig. 5(a). The KNN model alone achieved an accuracy of 0.9844, making it a highly competitive model, especially considering its minimal training time of just 0.0559 seconds. The model displayed strong classification performance across all classes, but class 4 had a slightly lower F1-score of 0.9674. The results indicate that KNN is an excellent choice for applications requiring fast model training while maintaining acceptable accuracy, as shown in Table 1 and confusion matrix given below in Fig. 5(b).

The SVM model alone achieved the highest accuracy among individual classifiers, reaching 0.9896 with a training time of 19.6371 seconds. The model performed exceptionally well across all classes,

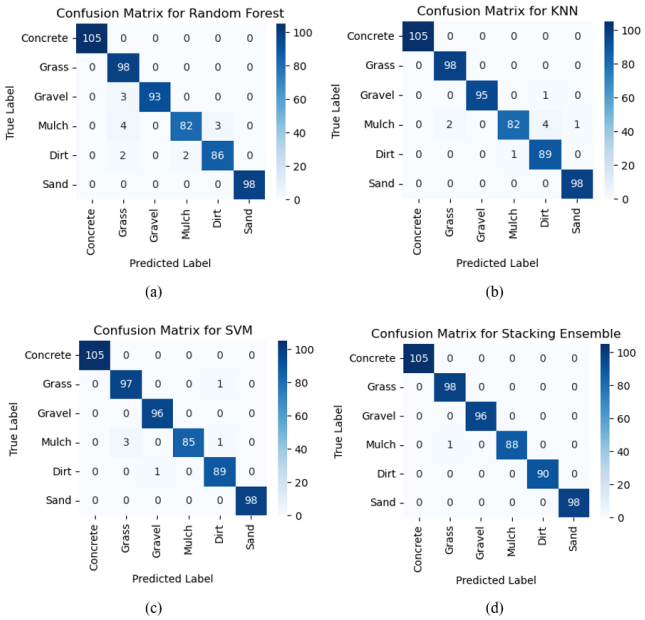

Figure 5: Confusion Matrix for (a) RF, (b) KNN, (c) SVM, and (d) proposed Stacking Ensemble model

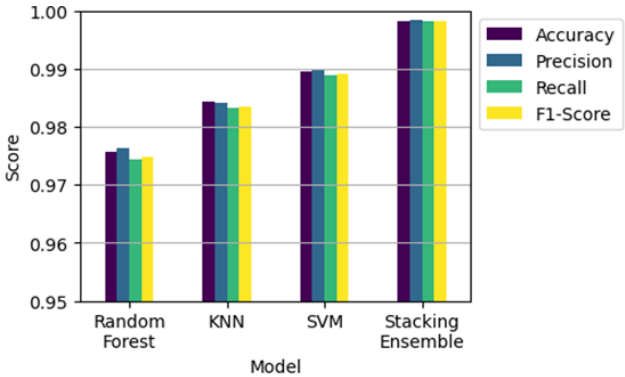

Figure 6: Comparison of all models for six classes

with a precision range of 0.9700 to 1.0000. The F1-score of 0.9892 confirms the model's high generalization ability. Class 3 had a slightly lower recall of 0.9551, but the overall classification performance was superior to Random Forest, given below in Table 1 and the confusion matrix in Fig. 5(c). The proposed SE model achieved the highest accuracy of 0.9983; however, it required significantly more training time (~316 seconds). It demonstrated better classification in most classes, with a precision and recall 1.0000 for four out of six categories. This result highlights the effectiveness of ensemble learning in improving model performance, which can be seen in the results below in Table 1 and the confusion matrix given in Fig. 5(d). Fig. 6 compares all models based on accuracy, precision, recall, and F1-score.

Table 1: Performance metrics for contrast (RF, KNN, SVM) and proposed SE model

Class	Precision				Recall				F1-Score			
	RF	KNN	SVM	SE	RF	KNN	SVM	SE	RF	KNN	SVM	SE
0	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.0000	1.000
1	0.915	0.980	0.970	0.989	1.000	1.000	0.989	1.000	0.956	0.989	0.979	0.995
2	1.000	1.000	0.989	1.000	0.968	0.989	1.000	1.000	0.984	0.995	0.995	1.000
3	0.976	0.988	1.000	1.000	0.921	0.921	0.955	0.988	0.948	0.953	0.977	0.994
4	0.966	0.946	0.978	1.000	0.955	0.989	0.989	1.000	0.961	0.967	0.983	1.000
5	1.000	0.989	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.995	1.000	1.000
Accuracy	0.975	0.984	0.989	**0.998**	-	-	-	-	-	-	-	-
Macro Avg.	0.976	0.984	0.989	**0.998**	0.974	0.983	0.989	**0.998**	0.975	0.983	0.98	**0.998**
Weighted Avg.	0.976	0.985	0.991	**0.998**	0.978	0.984	0.989	**0.998**	0.976	0.984	0.990	**0.998**
Train Time (s)	24.7	0.056	19.63	316.82	-	-	-	-	-	-	-	-

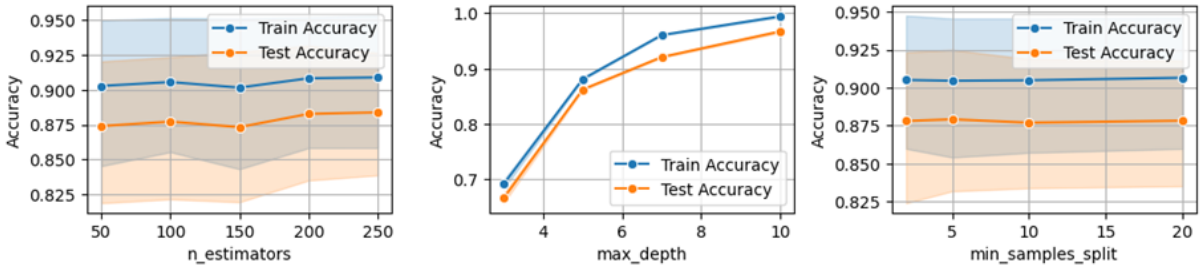

Figure 7: Hyperparameter tuning results for Random Forest: (a) *n_estimators*, **(b)** *max_depth* **and (c)** *min_samples_split*

5.2 Hyperparameter Tuning

Hyperparameter tuning was conducted for three machine learning models used in the proposed Stacked Ensemble Model: RF, SVM and KNN. The objective was to optimize test accuracy while balancing overfitting and generalization. The best-performing hyperparameter configurations for each model were identified through systematic tuning.

The best-performing Random Forest model used $n_estimators = 150$, $min_samples_split = 2$, and an unspecified max_depth, achieving a test accuracy of 0.9809, with a precision of 0.9818, recall of 0.9809, and F1-score of 0.9810. Increasing $n_estimators$ beyond 150 gave negligible gains (Fig. 7(a)), and restricting max_depth caused underfitting and reduced accuracy (Fig. 7(b)). Higher $min_samples_split$ lowered accuracy marginally; however, it helped reduce overfitting (Fig. 7(c)). The SVM model achieved the highest test accuracy of 0.9896 with RBF $kernel$, $C = 5.0$, and γ = scale, yielding a precision of 0.9897, recall of 0.9896, and F1-score of 0.9896. Increasing C improved flexibility but had diminishing returns (Fig. 8(a)). Linear and polynomial kernels performed well but were inferior to RBF (Fig. 8(b)). Sigmoid kernels consistently underperformed. Changing γ from *scale* to *auto* showed no effect on accuracy (Fig. 8(c)). KNN performed best with $metric = manhattan$, $n_neighbors = 3$, and $weights = distance$, achieving a test accuracy of 0.9844, precision of 0.9847, recall of 0.9844, and F1-score of 0.9843. Lower $n_neighbors \approx 3$–5 gave the best results (Fig. 9(a)), while larger values degraded performance. Distance weighting outperformed uniform (Fig. 9(b)), and $manhattan_metric$ slightly outperformed Euclidean and Minkowski (Fig. 9(c)), likely due to feature scaling.

5.3 Discussions

The results of our study demonstrated the effectiveness of ensemble machine learning techniques in multi-terrain classification for legged robots. The Stacking Ensemble model achieved the highest accuracy of 0.9983, outperforming the individual classifiers, including SVM (0.9896), KNN (0.9844), and Random Forest (0.9757). This section compares our findings with a few existing studies that utilized the QCAT dataset for terrain classification.

Zhao et al. [19] used SVM for terrain classification in a hexapod robot based on force and IMU features, enabling adaptive locomotion with a dynamic tripod gait. Their SVM achieved 96.7% accuracy, lower than the 99.83% achieved by our Stacking Ensemble, showing the strength of ensemble methods. A study by Kumar et al. [14] proposed stacked LSTM-based loss regularization for terrain classification, achieving an accuracy of 89% on the QCAT dataset. While their method improved efficiency by avoiding preprocessing variable-length sequences, our ensemble approach surpassed their classification accuracy. In another study, Ahmadi et al. [1] examined Gated Recurrent Neural Networks and validated their models on both PUTany and QCAT datasets. They observed an accuracy of around 97.63% for terrain classification, with their semi-supervised model handling real-world variations but requiring extensive training data. In contrast, our ensemble method achieved higher accuracy with lower computational complexity, offering a more practical solution for real-time robotic applications. Overall, our study contributes to the research in terrain classification by demonstrating that stacking ensemble methods can outperform both deep learning-based models and traditional clustering approaches.

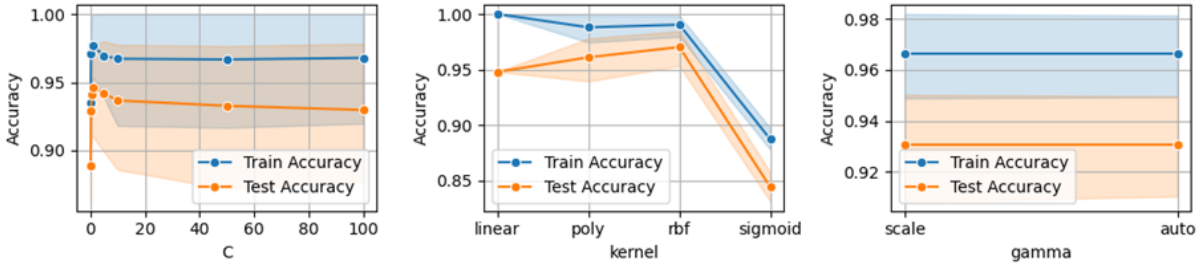

Figure 8: Hyperparameter tuning results for SVM: (a) C **, (b)** *kernel*, **and (c)** *gamma*

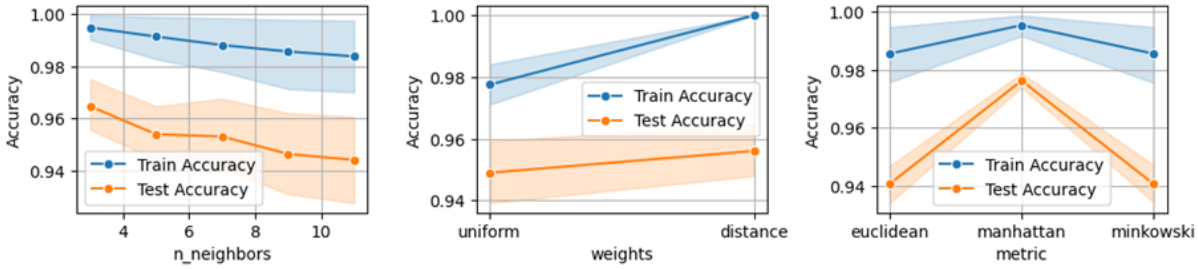

Figure 9: Hyperparameter tuning results for Random Forest (a) *n_neighbors*, **(b)** *weights* **and (c)** *metric*

6 CONCLUSIONS

This research has demonstrated the effectiveness of ensemble machine learning in improving multi-terrain classification for legged robots. By integrating RF, SVM, and KNN into a Stacking Ensemble, we achieved a superior accuracy of 0.998, showcasing the benefits of combining diverse learning models. However, this approach led to a longer training time (316.82s), which may pose challenges for real-time applications and thereby indicated the effectiveness of KNN with a trade-off between accuracy (0.984) and minimal training time (0.056s). Future work will optimize the ensemble approach for real-time deployment and expand the work with deep learning for various terrains, ensuring more adaptive mobility solutions.

REFERENCES

[1] Ahmadreza Ahmadi, Tønnes Nygaard, Navinda Kottege, David Howard, and Nicolas Hudson. 2021. Semi-supervised gated recurrent neural networks for robotic terrain classification. *IEEE Robotics and Automation Letters* 6, 2 (2021), 1848–1855.

[2] Reza Ahmadi, Tønnes Nygaard, Navinda Kottege, David Howard, and Nicolas Hudson. 2020. QCAT Legged Robot Terrain Classification Dataset. https://doi.org/10.25919/5f88b9c730442 Accessed: 2025-03-19.

[3] Rina Akabane and Yuka Kato. 2021. Pedestrian trajectory prediction based on transfer learning for human-following mobile robots. *IEEE Access* 9 (2021), 126172–126185.

[4] Stéphane Bazeille, Victor Barasuol, Michele Focchi, Ioannis Havoutis, Marco Frigerio, Jonas Buchli, Darwin G Caldwell, and Claudio Semini. 2014. Quadruped robot trotting over irregular terrain assisted by stereo-vision. *Intelligent Service Robotics* 7 (2014), 67–77.

[5] Leo Breiman. 2001. Random forests. *Machine learning* 45 (2001), 5–32.

[6] Iván del Pino, Angel Santamaria-Navarro, Anaís Garrell Zulueta, Fernando Torres, and Juan Andrade-Cetto. 2024. Probabilistic graph-based real-time ground segmentation for urban robotics. *IEEE Transactions on Intelligent Vehicles* 9, 5 (2024), 4989–5002.

[7] Emrah Dönmez and Adnan Fatih Kocamaz. 2020. Design of mobile robot control infrastructure based on decision trees and adaptive potential area methods. *Iranian Journal of Science and Technology, Transactions of Electrical Engineering* 44, 1 (2020), 431–448.

[8] Yanan Fan, Zhongcai Pei, Chen Wang, Meng Li, Zhiyong Tang, and Qinghua Liu. 2024. A review of quadruped robots: structure, control, and autonomous motion. *Advanced Intelligent Systems* 6, 6 (2024), 2300783.

[9] Ran Guo, Wenjiang Li, Yulong He, Tangjian Zeng, Bin Li, Guangkui Song, and Jing Qiu. 2024. Terrain slope parameter recognition for exoskeleton robot in urban multi-terrain environments. *Complex & Intelligent Systems* 10, 2 (2024), 3107–3118.

[10] Sehoon Ha, Joonho Lee, Michiel van de Panne, Zhaoming Xie, Wenhao Yu, and Majid Khadiv. 2024. Learning-based legged locomotion: State of the art and future perspectives. *The International Journal of Robotics Research* (2024), 02783649241312698.

[11] Md Mohsin Kabir, Jamin Rahman Jim, and Zoltán Istenes. 2025. Terrain detection and segmentation for autonomous vehicle navigation: A state-of-the-art systematic review. *Information Fusion* 113 (2025), 102644.

[12] Swapnil Saha Kotha, Nipa Akter, Sarafat Hussain Abhi, Sajal Kumar Das, Md Robiul Islam, Md Firoj Ali, Md Hafiz Ahamed, Md Manirul Islam, Subrata Kumar Sarker, Md Faisal Rahman Badal, et al. 2024. Next generation legged robot locomotion: A review on control techniques. *Heliyon* 10, 8 (2024), e37237.

[13] Oliver Kramer. 2013. K-Nearest Neighbors. In *Dimensionality Reduction with Unsupervised Nearest Neighbors*, Oliver Kramer (Ed.). Springer Berlin Heidelberg, Berlin, Heidelber, 13–23.

[14] Shakti Deo Kumar, Sudhanshu Tripathi, Krishna Ujjwal, Sarvada Sakshi Jha, and Suddhasil De. 2023. Robotic Terrain Classification with Loss Regularization. In *2023 3rd International Conference on Emerging Frontiers in Electrical and Electronic Technologies (ICEFEET)*. IEEE, Patna, India, 1–6.

[15] Ngum Breayon Kwe and Rashmi Priyadarshini. 2024. Emerging Trends in Mobile Robots. In *Robotics and Smart Autonomous Systems: Technology and Applications*, Rashmi Priyadarshini, Ram Mohan Mehra, Amit Sehgal, and Prabhu Jyot Singh (Eds.). CRC Press, Boca Raton, FL, 77–117.

[16] Derek A. Pisner and David M. Schnyer. 2020. Support vector machine. In *Machine Learning*, Andrea Mechelli and Sandra Vieira (Eds.). Academic Press, London, United Kingdom, 101–121.

[17] X Alice Wu, Tae Myung Huh, Rudranarayan Mukherjee, and Mark Cutkosky. 2016. Integrated ground reaction force sensing and terrain classification for small legged robots. *IEEE Robotics and Automation Letters* 1, 2 (2016), 1125–1132.

[18] Jialin Zhang, Jiamin Guo, Hui Chai, Qin Zhang, Yibin Li, Zhiying Wang, and Qifan Zhang. 2023. A day/night leader-following method based on adaptive federated filter for quadruped robots. *Biomimetics* 8, 1 (2023), 20.

[19] Yue Zhao, Feng Gao, Qiao Sun, and Yunpeng Yin. 2021. Terrain classification and adaptive locomotion for a hexapod robot Qingzhui. *Frontiers of Mechanical Engineering* 16, 2 (2021), 271–284.